

Lesson 1-4

Ethical Issues with AI

Learning Objectives

- Explain what algorithmic bias is and describe its main causes.
- Connect the basic principles of AI ethics (fairness, transparency, privacy protection, and accountability) to specific situations.

AI now helps make real decisions about people: selecting candidates for job interviews, identifying individuals through face recognition, and determining how personal data is used. But an AI can be unfair if the data it learned from was biased, and it can be hard to know why it made a decision at all. As AI spreads, questions arise about how to keep it fair, transparent, and accountable, and about who is responsible when it produces an incorrect or unfair result.

? Guiding question: What ethical issues arise as AI spreads, and what principles should guide how we use it?

✎ Explore · In Pairs

With a partner, look at this situation: a face-recognition AI is more likely to misidentify people from some ethnic groups than others. Before you read on, predict who could be harmed by this, and how. Give a reason.

Point!

1. Algorithmic Bias

(1) **(Algorithmic bias)**…bias in AI judgments caused by bias in the training data. *e.g., a hiring AI unfairly evaluating a particular gender; a face-recognition AI being more likely to misidentify particular ethnic groups*

(2) Main causes of bias

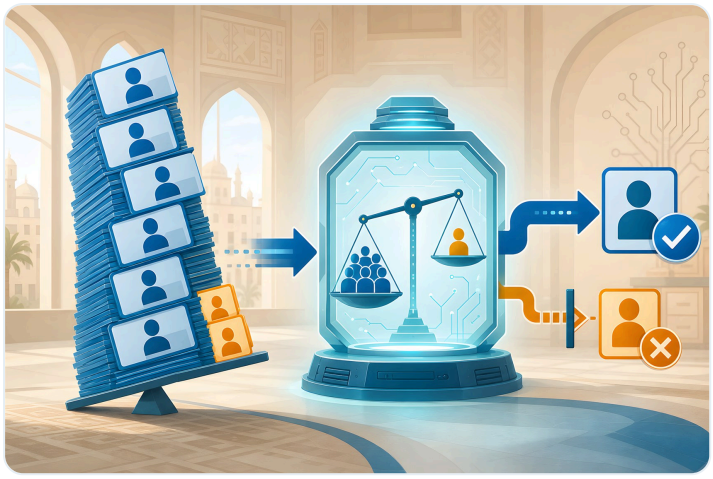
- Training data is biased (e.g., insufficient data on certain attributes; past discriminatory tendencies are reflected in the data). Bias can also enter through the way the system is built — for example, an inappropriate choice of variables, a proxy variable that indirectly stands in for a protected attribute, or the design of the model itself.

Key Fact:
If the training data is biased, an AI can reproduce that bias in its judgments; using AI responsibly means checking for bias, being able to explain decisions, and knowing who is accountable.

Key Concepts:

- algorithmic bias
- training data
- face recognition
- privacy
- explainable AI (XAI)
- black-box problem
- responsibility
- fairness
- transparency
- privacy protection
- accountability

KEY TERMS
algorithmic bias — bias in AI judgments caused by bias in the training data.



Biased training data leads to biased AI judgments

2. Privacy Issues

(1) The development of AI technology has given rise to new issues concerning **(privacy)**.

① Surveillance through face-recognition technology…cameras in public spaces can identify and track individuals.

② Mass collection of personal data…large amounts of online behavior data are collected and analyzed.

3. Explainable AI (XAI) and Responsibility

(1) **(Explainable AI (XAI))**…technology that makes it possible for humans to understand why an AI made a particular judgment.
When an AI's decision-making process is opaque (a black box), it is difficult to verify whether the result is correct.

(2) **(Responsibility)**…the question of who is responsible when an AI makes an incorrect judgment.
Views on responsibility differ depending on whether one is the developer, the user, the operator, etc.

4. Basic Principles of AI Ethics

(1) The following basic principles are considered important for using AI appropriately.

Principle	Description
Fairness	Not unjustly discriminating against any particular person or group
Transparency	Showing the AI's decision-making process and inner workings clearly
Privacy protection	Handling personal information appropriately and protecting privacy
Accountability	Being answerable for the AI's decisions

THINK IT THROUGH

When face-recognition technology is used in public spaces, safety and convenience may improve, but privacy concerns also arise. Explain how convenience, safety, and privacy protection should be balanced, and state which basic principle of AI ethics is most closely related to your view.



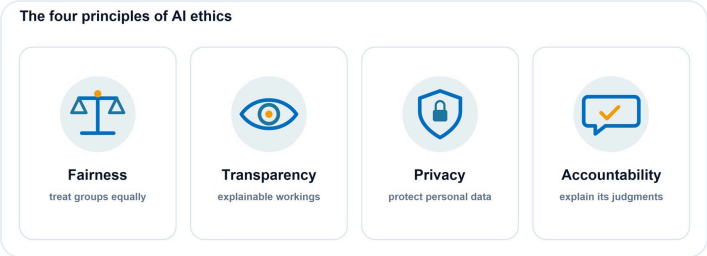
Privacy: control which personal data is shared.

KEY TERMS

explainable AI (XAI) — technology that makes it possible for humans to understand why an AI made a particular judgment.



Explainable AI reveals the reasons behind a decision.



The four principles of AI ethics

✎ Worked Example

(1) Mark each statement (A–D) with ○ if true or × if false.

A Algorithmic bias is bias in AI judgments caused by bias in the training data.

B Even if an AI's decision-making process is opaque, there is no problem as long as the result is correct.

C Explainable AI (XAI) is a technology that makes it possible for humans to understand the AI's reasoning.

D When an AI makes an incorrect judgment, who is responsible has already been clearly determined.

(2) Match each description (a–d) with the most appropriate AI ethics basic principle from the choices below (A–D).

a Not unjustly discriminating against any particular person or group

b Showing the AI's decision-making process and inner workings clearly

c Handling personal information appropriately and protecting privacy

d Being answerable for the AI's decisions

[Choices] **A** Fairness **B** Transparency **C** Privacy protection **D** Accountability

✓ Solution

(1)

A Correct definition of algorithmic bias. Therefore, ○.

B When an AI's decision-making process is opaque, it is difficult to verify whether the result is correct. Therefore, ×.

C Correct description of Explainable AI (XAI). Therefore, ○.

D Views on responsibility when an AI makes an incorrect judgment differ depending on whether one is the developer, the user, the operator, etc., and have not yet been clearly determined. Therefore, ×.

(2)

a: Not discriminating is fairness. **A**

PAUSE & THINK

If no one can explain why an AI rejected someone's job application, is that fair? Which principle of AI ethics is missing?

KEY TERMS

fairness — no unjust discrimination against a person or group.
accountability — being answerable for the AI's decisions.

b: Showing the decision-making process is transparency. **B**

c: Protecting privacy is privacy protection. **C**

d: Being answerable for decisions is accountability. **D**

Try

1 Answer the following questions.

1. What is the term for the phenomenon in which bias arises in AI judgments due to bias in the training data?
2. What is the term for the technology that makes it possible for humans to understand why an AI made a particular judgment? Also, give its three-letter abbreviation.
3. Among the basic principles of AI ethics, what is the term for the principle that requires not unjustly discriminating against any particular person or group?
4. Among the basic principles of AI ethics, what is the term for the principle that requires handling personal information appropriately?

2 Answer the following questions.

(1) From the following options (A–D), choose the one that is NOT an appropriate cause of algorithmic bias.

A Training data is biased.

B Past discriminatory tendencies are reflected in the data.

C The processing speed of the computer is slow.

D There is insufficient data on certain attributes.

(2) From the following options (A–D), choose all that are privacy issues associated with AI.

A Individuals are identified and tracked through face-recognition technology.

B Online behavior data is collected and analyzed in large quantities.

C AI creates creative works.

D AI computation speed improves.

Think as an Engineer — Investigate & Decide

Building on the hiring-AI case in the Think It Through note, investigate a real example and decide on a rule.

1 · Investigate. Find one documented case where an AI system was accused of being unfair or biased, and record the source you used (a news report, study, or official statement). What group did it affect, and what data might have caused it?

KEY TERMS

privacy — appropriate handling and protection of personal data.
responsibility — who answers when an AI judgment is wrong.

EXAM-STYLE QUESTION

Analyze how algorithmic bias can arise in an AI system, and how it can be reduced. **[6] In your answer:** refer to the cause of bias (biased or insufficient training data) and to how the principles of fairness and transparency help address it.

2 · Consider responsibility. When an AI makes an unfair decision, who is responsible — the developer, the company using it, or the operator? Note the responsibility each one holds.

3 · Decide. Before a company uses a hiring AI, what one rule should it follow to make the system fairer? Justify your rule using the principles of AI ethics.

Stuck? If the training data reflects past bias, the AI can repeat it — one fix is to check the data and test the AI's results across different groups.

🌍 In a New Context

A school wants to use face-recognition cameras at its gate to record attendance automatically. Identify one practical benefit for the school and one privacy concern. Then, if the camera misidentifies a student, who should be responsible — the school, or the company that made the AI? Give a reason.

❓ Lesson Question — Answered

The lesson question asked what ethical issues arise as AI spreads and what principles should guide how we use it. Because AI now makes decisions that affect people, three concerns stand out: an AI can be unfair when it learns from biased data; it can put people's privacy at risk; and it can be difficult to understand, or to hold anyone answerable for, its decisions. Handling these well takes more than accuracy. The lesson sets out four principles to aim for — fairness, transparency, privacy protection, and accountability — and keeping AI fair, transparent, privacy-respecting, and accountable is what makes its use responsible.

💡 THINK IT THROUGH

A hiring AI was found to rate applicants of a particular gender lower. Using the ideas of algorithmic bias and the basic principles of AI ethics, explain the most likely cause and one possible measure to address it. There can be more than one reasonable view. Give the basis for your answer.

KEY TERMS

transparency — showing the AI's decision-making process clearly.
privacy protection — handling personal information appropriately.

Exercise

1 Cover the Point! section with a red plastic sheet and quiz yourself in your notebook in numerical order.

2 Read the following passage and answer each question.

As AI technology spreads widely in society, attention has been drawn to various ethical issues. For example, there is the issue of (a), in which bias arises in AI judgments due to bias in the training data. Also, for the "black-box" problem—where it is unclear why an AI made a particular judgment—a technology called (

b) is being researched. Furthermore, when an AI makes an incorrect judgment, (c) is questioned, but no clear standard has yet been established among developers, users, and operators.

1. Fill in the blanks (a)–(c).

3 Answer the following questions.

(1) For each of a-d, indicate which AI ethics basic principle is most closely related: **A** Fairness, **B** Transparency, **C** Privacy protection, **D** Accountability.

a Ensuring that a hiring AI evaluates fairly regardless of gender.

b Disclosing the AI's decision-making process to users in an easy-to-understand way.

c Not using collected personal data for purposes other than as originally intended.

d Being answerable when an AI's diagnostic result turns out to be wrong.

(2) From the following options (A–D), choose the one that most appropriately describes ethical issues with AI.

A There are no ethical issues with AI.

B Algorithmic bias is caused by the AI's processing speed.

C Bias in training data can cause discriminatory bias in AI judgments.

D Even if an AI's decision-making process is opaque, there is no problem for users.

⭐ REFLECT & CHALLENGE

Reflect: Has this lesson changed how much you would trust an AI decision about you? Which of the four principles do you think is hardest to guarantee, and why?
Challenge: Choose one place AI is used (hiring, face recognition, or recommendations). Suggest one rule based on the AI ethics principles that would make it more trustworthy.

⭐ Key takeaway

Remember:
If training data is biased, an AI can reproduce that bias; using AI responsibly means checking for bias, being able to explain decisions, and knowing who is accountable — guided by fairness, transparency, privacy protection, and accountability.